TECHNICAL SHEET · PROJECT

Driver fatigue detection

Real-time fatigue and micro-sleep detection on a Raspberry Pi 5, with no cloud connection. Patent filed: OMPIC, 2025, no. 72322.

A prototype, not a certified safety system

This system is a prototype. It replaces neither the driver's attention nor approved safety equipment, and it measures nothing for medical purposes. The thresholds below are design parameters: the repository does not yet publish measured results (detection rate, frames per second).

5
indicators fused into one weighted score

2 s
of closed eyes triggers the micro-sleep alert

3 + 1
sound alert levels, including one priority alert

100%
of the processing runs locally

1. Processing chain

Stage	What happens
Sensors	Raspberry Pi camera module (or USB webcam) and a 3-pin heart-rate sensor.
Signals	dlib 68-point facial landmarks, then eye (EAR) and mouth (MAR) aspect ratios. Heart rate and variability (RMSSD).
Fusion	Weighted sum of five indicators, giving a fatigue score. A micro-sleep forces the score to the critical level.
Outputs	Buzzer with graded alerts, real-time dashboard, optional data log.

2. Indicators and thresholds

Indicator	Reference points	Weight
Blink rate	Normal: 12 to 20 per minute. Maximum fatigue below 5 or above 30.	35%
Heart rate	Normal: 60 to 100 BPM. Drowsiness below 50 BPM.	20%
Yawns (MAR)	MAR > 0.6. Normal: 0 or 1 in 5 minutes; high fatigue from 4.	20%
PERCLOS	Share of time with eyes closed over 1 minute. Alert below 8%; severe fatigue above 30%.	15%
Eye opening (EAR)	EAR < 0.21: eye closed. Wide open from 0.30.	10%
Micro-sleep	Eyes closed for 2 s or more with a face detected: the score rises to at least 80%.	Override

3. Alert levels

Level	Fatigue score	Buzzer pattern
Low	Above 25%	2 short beeps
Medium	From 40%	3 medium beeps
High	From 60%	5 rapid beeps
Micro-sleep	Priority	10 rapid beeps at full volume

4. Hardware and wiring

Component	Connection
Raspberry Pi 5	4 GB of memory or more recommended.
Pi Camera Module 2	Or a compatible USB webcam.
3-pin heart-rate sensor	VCC to 3.3 V (pin 1), GND to ground (pin 6), signal to GPIO 17 (pin 11).
Active buzzer	VCC to 5 V (pin 2), GND to ground (pin 14), signal to GPIO 18 (pin 12, PWM).

The repository does not yet contain a photo of the setup or a screenshot of the dashboard.

5. Software

Python, OpenCV, dlib, NumPy, SciPy, Flask for the dashboard, RPi.GPIO. A simulation mode runs the program without the hardware, and a headless mode prints results to the console. All thresholds and weights are set in config.py.

6. Known limitations

- No measured results published yet: detection rate, false alerts, frames per second.
- The thresholds are design values, to be tuned to the person, the camera and the lighting.
- A prototype that is not approved for road use.

7. Sources

All information comes from the public repository: github.com/yassinefartassi/driver-fatigue-detection. The weights, thresholds and alert levels are those in the code (modules/fatigue_fusion.py, config.py). Source code under the MIT licence. The dlib facial landmark model has its own terms (non-commercial use) and is not redistributed.